POLICY BRIEF · MALAWI · DISTRIBUTIONAL ANALYSIS

Who should be compensated when maize prices rise?

Allocating a fixed compensation budget when welfare losses differ across households and districts, and the government has imperfect information about poverty and market exposure.

Evidence: Malawi Sixth Integrated Household Survey (IHS6, 2024/25), 12,502 households. Official poverty statistics reproduced exactly before any simulation. Validated run `maize_20261025T173130_1dafdd`. October 2026.

Key messages

1. **A maize price rise is regressive.** In a scenario where retail grain prices rise 25%, the poorest tenth of the population loses 4.1% of its consumption; the richest tenth is unaffected. Half the population lives in losing households. National poverty rises from 47.3% to 48.7% and ultra-poverty from 24.0% to 26.0%.
2. **How the money is targeted matters more than how much is known about maize markets.** With the same budget, a well-specified proxy means test (PMT) achieves about 75% of what perfect information could, against 48% for an equal per-person transfer and 54% for district-based allocation. Adding information on who buys maize does not measurably improve on a good poverty proxy.
3. **Choosing the objective is a policy decision, not a technical one.** Filling the deepest poverty gaps first and reimbursing the losses of poor households reduce the poverty gap equally, but differ sharply on who is protected: the first compensates only 14% of poor households' losses, the second 99%.
4. **District targeting alone cannot reach most of the exposed.** Districts explain only 6–11% of the variation in households' losses and post-shock poverty; most of the need is *within* districts.
5. **The PMT figures are an upper bound.** They assume registry data as accurate and current as the survey itself. Real registries are older and noisier, so the realistic advantage of PMT targeting is smaller than shown.

1. The policy problem

Maize is Malawi's staple: about three-quarters of households grow it, yet in the week before the IHS6 interview 50% of the population lived in households that had bought it. Prices are volatile. Between June 2024 and March 2025 IFPRI recorded retail maize rising from about K600 to K1,700 per kg. When prices rise, households that sell more maize than they buy gain; net buyers lose. Governments facing such shocks typically have a fixed budget for compensation and must decide *who* receives it with imperfect data on household poverty and market exposure.

This brief compares realistic compensation rules at an **identical budget**, set equal to half of the aggregate losses of losing households, against benchmarks that assume perfect knowledge of every household. It reports which rule performs best under three different policy objectives, and how confident we can be in the ranking.

How the analysis works

- **Welfare impact.** For each household, the first-order money-metric loss from a price rise equals purchases × retail price change, minus sales × farm-gate price change. Retail and farm-gate prices are allowed to move differently, and maize the household grows and eats has no first-order effect.
- **Price changes are scenarios, not forecasts.** The central scenario raises retail grain prices by 25%, flour by 21% and farm-gate prices by 12.5%. Results were tested across 972 combinations of price change, transmission, flour pass-through, demand response, budget size and measurement choices.
- **Uncertainty** comes from a complex-survey bootstrap (200 replicates that resample enumeration areas within strata). In every replicate the government's targeting models are re-estimated, so the uncertainty in its own information is included.

2. Who loses: distributional findings

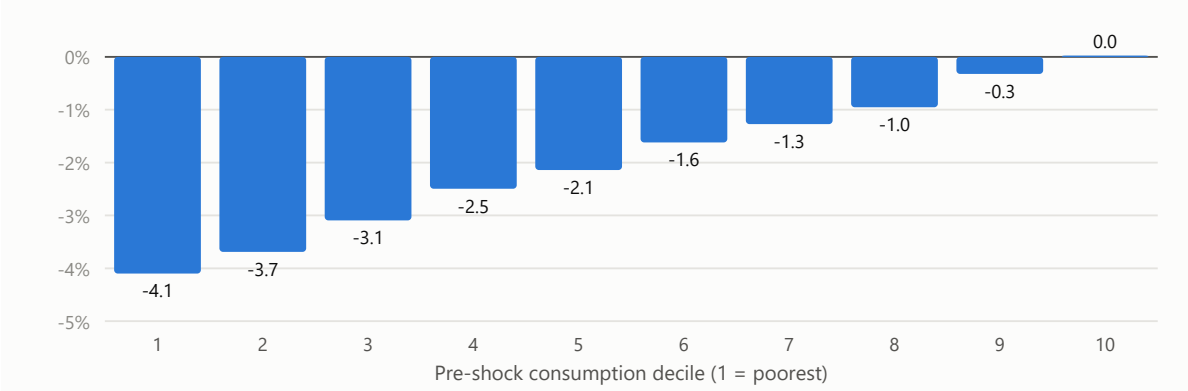


Figure 1. Welfare change as % of consumption, by pre-shock consumption decile (central scenario, no compensation). Losses fall steadily from −4.1% in the poorest decile to roughly zero in the richest.

The poor are disproportionately net buyers. 67% of people in the poorest decile live in households that bought maize and sold none, against 23% in the richest decile. Sellers are concentrated higher up the distribution (6% of the poorest decile vs 29% of the richest), so gains from higher farm-gate prices accrue mainly to the better-off.

Aggregate effects are moderate; distributional effects are not. The net loss is about 1% of national consumption, but the poverty gap rises from 15.7 to 16.7 and the squared poverty gap, which weights the poorest most, from 7.1 to 7.8.

Market position (share of population)	%
Bought maize in past 7 days, sold none	44.5
Sold maize in season, no recent purchase	13.7
Both bought and sold	6.0
Neither (in reference periods)	35.8

A 7-day purchase window understates buying: the share of households buying ranges from 33% (May, after harvest) to 71% (February, lean season).

3. Compensation choices at the same budget

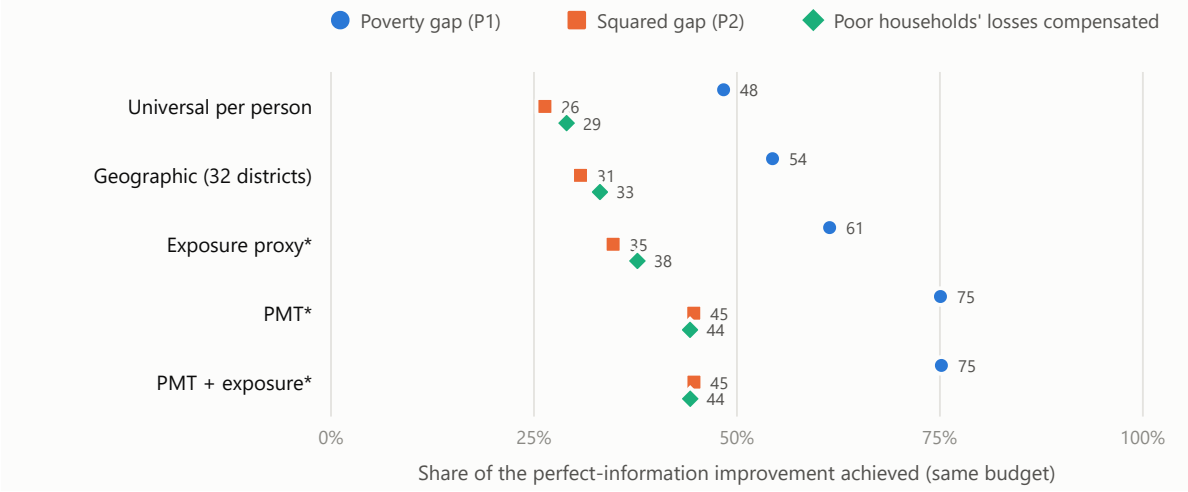


Figure 2. Share of the improvement a perfect-information rule would achieve, for each rule and objective (central scenario, identical budget). * Survey-ideal: estimated on the same survey round, so an upper bound for a real registry.

Rule (same budget)	What the government must know	Poverty gap	Squared gap	Losses of poor compensated
No compensation	–	16.69	7.76	0%
Equal per-person transfer	Population counts	16.23	7.44	29%
Geographic (32 districts)	District survey estimates	16.17	7.39	33%
Exposure proxy	Household observables survey-ideal	16.11	7.34	37%
Poverty proxy (PMT)	Household observables survey-ideal	15.98	7.23	44%
PMT + exposure	Household observables survey-ideal	15.97	7.23	44%
Benchmark: fill deepest gaps first	*Every household's true consumption (infeasible)*	*15.74*	*6.57*	*14%*
Benchmark: reimburse poor households' losses	*Every household's true loss (infeasible)*	*15.74*	*7.12*	*99%*

Poverty gap and squared gap are ×100 and person-weighted (lower is better). "Losses of poor compensated" is the share of the value of shock losses borne by initially poor households that transfers offset (higher is better).

What the comparison shows

- **PMT-type rules come first on every objective**, and in every one of the 972 scenarios tested. The equal per-person transfer is always last among implementable rules; the geographic rule sits between the two.
- **Adding information on who buys maize to the PMT changes nothing measurable.** The difference is within ±0.003 on every objective, with confidence intervals spanning zero. Poverty and maize-buying are strongly correlated, so a good poverty proxy already reaches most poor losers.
- **The two perfect-information benchmarks reduce the poverty gap equally but protect different people.** The poverty-gap measure counts any kwacha given to anyone below the line equally, so it cannot distinguish between them. If the policy goal is to protect households hurt by the shock, the objective must say so explicitly; if it is to help the very poorest first, the squared poverty gap is the relevant measure.

4. Geographic implications

District (highest post-shock poverty gap)	Post-shock gap	P(in top 8)
Mulanje	32.7	1.00
Phalombe	27.3	0.95
Nsanje	25.9	0.93
Zomba Non-City	23.6	0.54
Chikwawa	23.5	0.65
Blantyre (rural)	23.2	0.65
Machinga	23.2	0.64
Thyolo	22.9	0.44

P(in top 8): share of bootstrap replicates in which the district ranks among the 8 with the highest post-shock poverty gap.

The Southern Region dominates the ranking, but only three districts are clear priorities. Mulanje, Phalombe and Nsanje rank in the top eight in at least 93% of replicates. For the remaining priority places, survey precision cannot separate one district from the next.

Exposure and depth of poverty do not coincide. Some districts with the largest proportional losses, such as Salima (−4.0% of consumption), are not among the poorest.

Most need is within districts. Districts explain 6–11% of the household variation in losses and post-shock poverty. A district formula can therefore improve only modestly on an equal transfer.

Finer geographic targeting may do much better. Recent World Bank work for Malawi (Gualavisi & Newhouse, *WBER* 2025) finds that village-level predictions using a partial registry and satellite data rank village welfare far more accurately than existing registry scores. That approach was not tested here.

5. How certain are these results?

- **Levels are uncertain; rankings are not.** The post-transfer poverty gap under any rule has a standard error of about 0.44 points, but the *differences* between rules are estimated precisely. Compared with PMT, the equal transfer raises the poverty gap by 0.25 (95% CI 0.24–0.27), the geographic rule by 0.19 (0.18–0.21) and the exposure proxy by 0.14 (0.12–0.16).
- **The government's own estimation error matters little.** Re-estimating the targeting models in each replicate barely changes the results. PMT and PMT + exposure split the "best rule" position (62% vs 38% of replicates on the poverty gap); no other rule is ever best.
- **The distance from perfect information is clear:** about 0.24 points of the poverty gap (95% CI 0.22–0.26), a quarter of the improvement that is attainable in principle.
- **Assumptions change the size of the shock, not the ranking.** Flour pass-through moves the poverty impact by about 45%, farm-gate transmission by about 8%, and seasonal adjustment of purchases by about 3%.

6. Implications for compensation design

- **State the objective before choosing the instrument.** "Reduce poverty", "protect the poorest" and "compensate those hurt by the shock" lead to different allocations even with perfect data.
- **Invest in household-level poverty information before maize-market information.** In this analysis, better poverty targeting is worth far more than knowing who buys maize. The remaining gap to perfect information is a measure of the value of better household data.
- **Treat district formulas as a complement, not a substitute.** They are transparent and cheap but cannot reach the within-district variation where most need lies. Use them alongside household-level targeting or finer geography.
- **Discount PMT performance for registry realism.** The PMT advantage shown here is an upper bound. Registry data collected years earlier, or with enumeration error, will narrow it. Under some conditions, simpler rules have been found to perform as well (Brown, Ravallion & van de Walle, 2018).

7. Limitations

- **Scenario, not forecast or causal estimate.** Price changes are assumed. The model holds the harvest fixed and excludes wage, labour, supply-response and general-equilibrium effects.
- **Upper-bound targeting.** Proxy-means-test and exposure models are estimated and evaluated on the same 2024/25 survey. Registry staleness, misreporting and partial coverage are not modelled.
- **Noisy measures of household exposure.** Purchases come from 7-day recall, annualised, with strong seasonality. Fieldwork coincided with an extreme maize price spike, but purchases are deflated with the general consumer price index rather than a maize-specific price. About a third of sales records predate the fieldwork window.
- **District geography only.** No sub-district estimates or maps: census microdata were not accessed.
- **Uncertainty is design-based but incomplete.** It excludes re-calibration of survey weights, finite-population corrections and measurement error.

Data and methods in brief

Malawi IHS6 2024/25 (NSO; World Bank Microdata Library MWI_2024-2025_IHS-VI_v01_M), 12,502 households, 782 enumeration areas, released strata and weights. Official IHS5 and IHS6 poverty statistics, including all 32 IHS6 district estimates, were reproduced exactly before any simulation. Welfare: real per-capita consumption (July 2024 prices) from the official aggregate. Targeting models: weighted least squares on household demographics, dwelling, assets and land, cross-fitted by enumeration area so that no household is scored by a model trained on its own area. Uncertainty: Rao–Wu rescaled bootstrap. Every run produces a manifest of code and input hashes, parameters, seeds and package versions, and passes 10 synthetic-data tests before touching the survey data.

Sources: NSO (2026) IHS6 Monetary Poverty Findings; Chiwaula et al. (2024) IFPRI MaSSP WP45; Gualavisi & Newhouse (2025) *World Bank Economic Review* 39(2); Brown, Ravallion & van de Walle (2018) *Journal of Development Economics* 134; Fuje & Pullabhotla (2020) World Bank PRWP 9436; Deaton (1989) *Economic Journal* 99; Araar & Verme, SUBSIM. Figures are simulations under stated assumptions using survey microdata. They do not identify any community or household.